

Business Requirement Document

Returns & Refunds Module

ShopSphere E-Commerce Platform — Platform Modernization Program (FY2026)

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Author	Vikram Singh, Senior Business Analyst
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1. Document Information

Program	ShopSphere Platform Modernization Program (FY2026)
Module	Returns & Refunds
Prepared For	Rajesh Mehta, VP of E-Commerce (Executive Sponsor)
Prepared By	Vikram Singh, Senior Business Analyst
Distribution	Product, IT/Architecture, QA, Risk & Compliance, Business teams

2. Executive Summary

The ShopSphere E-Commerce Platform serves approximately 2 million monthly active users across India, UAE, and Southeast Asia, generating substantial return and refund volumes that directly influence customer trust, repeat purchase rate, and Gross Merchandise Value (GMV) retention. The current returns process is fragmented across manual customer support workflows, inconsistent category-level return windows, and delayed refund disbursement, resulting in elevated customer complaints and reconciliation overhead for Finance. This Business Requirement Document defines the requirements for a modernized Returns & Refunds Module that will digitize return initiation, enforce category-specific return policy windows, orchestrate reverse pickup through logistics partners, standardize the quality check (QC) workflow, and automate refund processing across original payment method, wallet credit, and bank transfer channels. The module will integrate with PayEdge Payment Gateway, SAP S/4HANA, the Inventory Management System (IMS), Salesforce CRM, and SwiftShip Logistics to deliver an end-to-end, auditable, and SLA-driven returns experience. This initiative is expected to reduce average refund turnaround time, curb return-related fraud through serial-returner detection, and improve Net Promoter Score (NPS) for post-purchase experience.

3. Business Objective

The primary business objective is to deliver a self-service, policy-compliant, and financially reconciled Returns & Refunds capability that reduces manual intervention, shortens refund SLA, and minimizes revenue leakage from return fraud and mis-restocked inventory. The module must enforce differentiated return windows per product category (e.g., electronics vs. apparel), provide transparent RMA (Return Merchandise Authorization) tracking to customers, and ensure that refunds are only disbursed after a successful QC outcome, except where policy explicitly permits refund-without-return. Secondary objectives include reducing the cost-to-serve for customer support agents handling return escalations, improving inventory recovery rates through faster restocking cycles, and providing Compliance with an auditable trail of every return and refund transaction for regulatory and tax reporting purposes across India, UAE, and Southeast Asia.

4. Project Scope

In Scope

- Return initiation by customers via web and mobile app, including RMA generation.
- Return reason capture with structured, category-aware reason codes.
- Reverse pickup scheduling with logistics partners, primarily SwiftShip Logistics.
- Quality check (QC) workflow upon receipt of returned items at the warehouse.
- Refund processing via original payment method, wallet credit, or bank transfer.
- Product exchange handling (size, color, or SKU substitution) as an alternative to refund.
- Enforcement of the return policy window per product category.

Out of Scope

- Warranty claims for third-party brand manufacturers (handled directly by brand OEM support channels).
- Payment card network dispute and chargeback management, which remains within the Finance team's separate chargeback process.
- Reverse logistics network design, including hub placement and carrier contract negotiation.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Vikram Singh	Senior Business Analyst	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Regulatory compliance

6. Business Requirements

ID	Business Requirement
BR-001	The platform shall enable customers to initiate a return request independently through web and mobile channels without mandatory contact with customer support.
BR-002	The platform shall enforce return eligibility windows that vary by product category to align with vendor agreements and product risk profiles.
BR-003	The platform shall provide a standardized quality check process to determine refund eligibility of returned merchandise before funds are disbursed.
BR-004	The platform shall support multiple refund disbursement methods to accommodate customer payment preferences and regional banking constraints, including COD orders.
BR-005	The platform shall allow customers to opt for a product exchange in lieu of a monetary refund where inventory availability (ATP) permits.
BR-006	The platform shall detect and flag serial-returner fraud patterns to protect margin and reduce abuse of the return policy.
BR-007	The platform shall reconcile every completed return with SAP S/4HANA through credit note and reverse invoice postings for financial accuracy.
BR-008	The platform shall provide customer service representatives with a unified view of return case status to reduce resolution time for escalations.

7. Functional Requirements

Return Initiation

- **FR-001** The system shall allow a customer to select an eligible delivered order and item(s) to initiate a return request from the "My Orders" section.
- **FR-002** The system shall present a structured list of return reason codes (e.g., "Defective/Damaged", "Wrong Item Delivered", "Size/Fit Issue", "No Longer Needed", "Better Price Available") specific to the item's category.

- **FR-003** The system shall generate a unique RMA (Return Merchandise Authorization) number immediately upon successful submission of a return request and display it along with expected pickup timelines.
- **FR-004** The system shall validate the return request against the category-specific return policy window at the time of submission and reject requests outside the eligible window with a clear reason message.
- **FR-005** The system shall require photo evidence upload for return reasons of "Defective/Damaged" or "Wrong Item Delivered" before allowing RMA generation.

Pickup & QC

- **FR-006** The system shall integrate with SwiftShip Logistics to schedule a reverse pickup slot and communicate the scheduled date/time window to the customer.
- **FR-007** The system shall update the return case status through defined states: RMA Created, Pickup Scheduled, Picked Up, In Transit, Received at Warehouse, QC In Progress, QC Passed/Failed, Refund/Exchange Initiated, Closed.
- **FR-008** The system shall provide a QC workflow interface for warehouse staff to record QC outcome (Pass/Fail), condition notes, and mismatched-item flags upon physical receipt of the returned product.
- **FR-009** The system shall route QC-failed items to a rejection workflow that notifies the customer with the failure reason and applicable next steps (e.g., item return to customer, partial refund, or dispute escalation).

Refund Processing

- **FR-010** The system shall automatically trigger refund processing through PayEdge Payment Gateway to the original payment method upon a QC Pass outcome, for eligible payment types.
- **FR-011** The system shall support refund to ShopSphere Wallet credit as an alternative, offering an incentive credit percentage configurable by Product for wallet-based refunds.
- **FR-012** The system shall support refund via direct bank transfer (NEFT/IMPS or regional equivalent) for COD orders where no original digital payment method exists.
- **FR-013** The system shall generate a corresponding credit note and reverse invoice in SAP S/4HANA for every completed refund transaction.

Exchanges

- **FR-014** The system shall allow customers to select an alternate size, color, or SKU variant as an exchange option, subject to real-time ATP (Available to Promise) inventory check.
- **FR-015** The system shall dispatch the replacement item only after QC Pass confirmation of the originally returned item, unless the item qualifies for the reverse-pickup-free "swap-at-doorstep" exchange flow.
- **FR-016** The system shall update NetSuite WMS and the internal Inventory Management System (IMS) to restock QC-passed items into sellable inventory within the applicable SLA.

8. Non-Functional Requirements

Performance

- **NFR-001** Return initiation and RMA generation API calls shall respond within 2 seconds at the 95th percentile under normal load.
- **NFR-002** The QC workflow interface shall load item and order history within 2 seconds for warehouse operators.

Security

- **NFR-003** All refund transactions shall comply with PCI-DSS requirements when interfacing with PayEdge Payment Gateway.
- **NFR-004** Customer PII and bank account details captured for bank transfer refunds shall be encrypted using AES-256 at rest and TLS 1.2+ in transit.
- **NFR-005** The module shall be compliant with OWASP Top 10 security controls, validated through periodic penetration testing.

Availability

- **NFR-006** The Returns & Refunds Module shall maintain 99.9% uptime for customer-facing return initiation services.

Scalability

- **NFR-007** The module shall scale to handle up to 6x normal return volume during peak periods following major sale events (e.g., end-of-season sales).

Reliability

- **NFR-008** Critical return and refund services shall have a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour.
- **NFR-009** Refund transaction records shall not be lost or duplicated in the event of a mid-transaction system failure, enforced through idempotent transaction handling.

9. User Stories

- **US-001** As a customer, I want to initiate a return directly from my order history, so that I do not need to contact customer support for a straightforward return.
- **US-002** As a customer, I want to see the applicable return window for each product category before purchase, so that I can make an informed buying decision.
- **US-003** As a customer, I want to track my RMA status in real time, so that I know when my refund or exchange will be processed.
- **US-004** As a warehouse QC operator, I want a guided checklist for quality checking returned items, so that I can consistently determine refund eligibility.
- **US-005** As a customer, I want to choose between refund to my original payment method, wallet credit, or bank transfer, so that I receive my money in the way that suits me best.
- **US-006** As a Compliance Officer, I want every refund transaction to be logged with supporting QC and payment evidence, so that we remain audit-ready for regulatory review.
- **US-007** As a customer service representative, I want a single dashboard view of a customer's return case history, so that I can resolve escalations faster.
- **US-008** As a Fraud Analyst, I want automated flags on customers with abnormally high return rates, so that I can investigate potential return fraud.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given a delivered order within the return window, when the customer selects "Return Item", then the return initiation flow shall launch with eligible items listed.	Return flow displayed with correct eligible items
AC-002	FR-003	Given a valid return request submission, when the customer confirms the reason and item, then a unique RMA number shall be generated and displayed.	RMA number generated and visible within 2 seconds
AC-003	FR-004	Given an electronics item purchased 10 days ago, when the customer attempts a return, then the system shall reject the request as outside the 7-day window.	Return request rejected with policy message
AC-004	FR-006	Given an approved RMA, when pickup scheduling is triggered, then SwiftShip Logistics shall return a confirmed pickup slot within 24 hours.	Pickup slot confirmed and communicated to customer
AC-005	FR-008	Given a returned item received at the warehouse, when the QC operator records a Pass outcome, then the case status shall move to "Refund/Exchange Initiated".	Status transition triggers refund/exchange workflow
AC-006	FR-009	Given a QC Fail outcome, when the failure is recorded, then the customer shall receive a notification with the failure reason within 1 hour.	Customer notified with clear rejection reason
AC-007	FR-010	Given a QC Pass on an item paid via card, when refund processing triggers, then PayEdge Payment Gateway shall disburse the refund within the defined SLA.	Refund credited to original payment method within SLA
AC-008	FR-013	Given a completed refund transaction, when the refund is finalized, then a credit note shall be posted in SAP S/4HANA.	Credit note and reverse invoice recorded in ERP
AC-009	US-008	Given a customer with returns exceeding the fraud threshold in a rolling 90-day period, when a new return is submitted, then the system shall flag the case for manual fraud review.	Case routed to fraud review queue before RMA approval

11. Business Rules

- **RULE-001** Electronics category items are eligible for return within 7 days of delivery, provided original packaging and accessories are intact.
- **RULE-002** Apparel and footwear category items are eligible for return within 15 days of delivery.
- **RULE-003** Non-returnable categories (e.g., innerwear, perishable goods, personalized/customized products, digital gift cards) shall be explicitly marked non-returnable on the PDP and blocked from return initiation.
- **RULE-004** Refunds shall be disbursed only after a QC Pass outcome, except for pre-approved "refund-without-return" cases for low-value items below a configurable threshold.
- **RULE-005** Wallet credit refunds may include an incentive bonus of up to 5% over the original order value, configurable by Product.
- **RULE-006** A customer flagged with more than 5 returns in a rolling 90-day period, or a return rate exceeding 40% of total orders, shall be routed to manual fraud review before RMA approval.
- **RULE-007** Exchange requests are permitted only when the replacement SKU variant has positive ATP inventory at the time of request confirmation.
- **RULE-008** COD orders are not eligible for refund to "original payment method" and shall default to wallet credit or bank transfer.

12. Process Flow Description

Primary Flow — Return to Refund:

1. Customer selects a delivered order item and initiates a return request via web or mobile app.
2. System validates the item's category-specific return policy window (RULE-001, RULE-002) and eligibility (RULE-003).
3. Customer selects a return reason and, if required, uploads photo evidence (FR-005).
4. System generates an RMA number and records the case in Salesforce CRM.
5. System schedules a reverse pickup with SwiftShip Logistics and notifies the customer of the pickup window via SendGrid Email API and Twilio SMS Gateway.
6. Logistics partner collects the item and updates status to "Picked Up" then "In Transit".
7. Item arrives at the designated warehouse; status updates to "Received at Warehouse".
8. Warehouse QC operator inspects the item against the QC checklist and records a Pass or Fail outcome.
9. On QC Pass, the system triggers refund processing (via PayEdge Payment Gateway, wallet credit, or bank transfer) or initiates the exchange dispatch, per customer selection.
10. System posts a credit note and reverse invoice to SAP S/4HANA and updates NetSuite WMS/IMS to restock the item.
11. Customer receives confirmation of refund credit or exchange dispatch; case status is set to "Closed".

Alternate Flow — QC Failure:

1. Warehouse QC operator records a QC Fail outcome with a documented reason (e.g., item damaged by customer, missing accessories, counterfeit suspicion).
2. System halts refund/exchange processing and notifies the customer of the rejection with supporting reason and photos where applicable.
3. Customer is offered the option to have the item shipped back at their cost, dispute the QC finding through customer support, or accept the rejection.
4. If disputed, the case is escalated to a Customer Service Representative via Salesforce CRM for manual review with the QC evidence.
5. Final resolution (rejection upheld, partial refund, or override to Pass) is recorded, and the case is closed with an audit trail entry.

13. Assumptions

- All product categories are pre-tagged with an applicable return policy window in the product catalog prior to module go-live.
- SwiftShip Logistics has available reverse pickup capacity across all serviceable pin codes/postal areas in India, UAE, and Southeast Asia.
- PayEdge Payment Gateway supports refund-to-source transactions for all payment methods accepted at checkout.

- Warehouse staff will be trained on the standardized QC checklist prior to rollout.
- Customers have access to a registered mobile number and email for OTP-based verification and status notifications.
- Finance has approved the wallet credit incentive funding model referenced in RULE-005.

14. Constraints

- Refund processing timelines are dependent on external banking and payment network cutoffs beyond ShopSphere's direct control.
- Reverse pickup availability is constrained by SwiftShip Logistics' serviceable geographic coverage.
- Regulatory refund timelines and consumer protection rules differ across India, UAE, and Southeast Asia markets, requiring region-specific configuration.
- Legacy order records migrated prior to the modernization program may lack structured category metadata needed for automated policy window enforcement.

15. Dependencies

- Dependency on the Catalog/PDP module for accurate, up-to-date category and return-window metadata.
- Dependency on PayEdge Payment Gateway for refund disbursement capability and transaction status callbacks.
- Dependency on SAP S/4HANA for credit note and reverse invoice posting.
- Dependency on NetSuite WMS and the internal Inventory Management System (IMS) for restocking and ATP validation for exchanges.
- Dependency on Salesforce CRM for return case tracking and customer communication history.
- Dependency on SwiftShip Logistics API for reverse pickup scheduling and shipment tracking.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Delayed reverse pickup by logistics partner extends overall refund SLA and harms customer satisfaction.	Medium	High	Define contractual pickup SLA with SwiftShip Logistics and monitor via dashboard alerts.
RISK-002	Inconsistent QC judgment across warehouse operators leads to disputed refund rejections.	Medium	Medium	Standardize QC checklist and conduct periodic QC audit sampling.
RISK-003	Serial-returner fraud increases return-related margin erosion during high-sale periods.	High	High	Implement automated fraud-pattern detection with configurable thresholds (RULE-006).
RISK-004	PayEdge Payment Gateway outage delays refund disbursement beyond committed SLA.	Low	High	Implement retry queue with failover notification and manual reconciliation fallback.
RISK-005	Incorrect ERP credit note posting causes financial reconciliation discrepancies.	Medium	High	Automated reconciliation job between Returns Module and SAP S/4HANA with daily variance reporting.
RISK-006	Peak return volume post major sale events (up to 6x normal) overwhelms QC workflow capacity.	Medium	Medium	Scale warehouse QC staffing and system throughput ahead of known sale calendars.

17. Data Requirements

The Returns & Refunds Module shall capture and persist structured data across the return lifecycle to support customer transparency, financial reconciliation, and compliance audit trails. Return and refund records shall be retained for a minimum of 7 years to satisfy financial audit and regional tax regulations, while return-reason and QC evidence data shall be retained for 2 years for fraud analytics purposes.

Data Field	Description
RMA ID	Unique identifier generated per return request
Order ID / SKU	Reference to the original order and item(s) being returned
Return Reason Code	Structured reason selected by the customer
QC Status	Pass, Fail, or Pending, with operator notes
Pickup Schedule Details	Scheduled date/time window and logistics partner reference ID

Refund Method	Original payment method, wallet credit, or bank transfer
Refund Amount	Monetary value disbursed, including any wallet incentive bonus
Exchange SKU	Replacement SKU variant selected, where applicable
Case Status Timeline	Timestamped history of all RMA status transitions

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
PayEdge Payment Gateway	Refund disbursement to original payment method	Outbound/Inbound	Refund request, transaction status, disbursement confirmation
SAP S/4HANA (ERP)	Credit note and reverse invoice posting	Outbound	RMA ID, refund amount, tax details, order reference
NetSuite WMS / Inventory Management System (IMS)	Restocking of QC-passed items and ATP validation for exchanges	Bidirectional	SKU, QC status, restock quantity, ATP availability
Salesforce CRM	Return case tracking and customer communication history	Bidirectional	RMA ID, case status, customer interaction logs
SwiftShip Logistics	Reverse pickup scheduling and shipment tracking	Bidirectional	Pickup address, scheduled slot, shipment tracking status
SendGrid Email API / Twilio SMS Gateway	Return status and refund notifications to customers	Outbound	RMA status updates, refund confirmation messages

19. Reporting Requirements

- Daily returns volume dashboard segmented by product category and region.
- Refund turnaround time (TAT) report tracking SLA adherence from QC Pass to disbursement.
- QC pass/fail rate report by warehouse location and category.
- Serial-returner fraud monitoring report with flagged customer accounts and case outcomes.
- Return reason trend analysis to inform product quality and listing accuracy improvements.
- Finance reconciliation report comparing refund disbursements to ERP credit note postings.
- Exchange fulfillment report tracking ATP availability impact on exchange completion rate.

20. Success Metrics

KPI	Target
Average refund turnaround time (QC Pass to disbursement)	Less than 48 hours for digital refunds; less than 5 business days for bank transfer
Return rate as % of total orders	Reduce by 10% year-over-year through improved PDP accuracy and reason analytics
QC-to-refund SLA adherence	95% of cases processed within defined SLA
Customer satisfaction (CSAT) on return experience	Achieve 4.3/5.0 or higher
Fraud detection accuracy (serial returners)	85% precision in flagged fraud cases upon manual review
Restocking cycle time (QC Pass to sellable inventory)	Under 24 hours at 95th percentile

21. Future Enhancements

- AI-based automated QC assessment using image recognition to reduce manual inspection time.
- Predictive return-likelihood scoring at the point of purchase to proactively manage inventory and customer messaging.
- Instant refund feature for trusted customers based on historical return behavior, bypassing wait-for-QC for select categories.
- Self-service doorstep exchange without requiring reverse pickup, expanded to additional categories beyond current "swap-at-doorstep" pilot.
- Integration with third-party resale/liquidation marketplaces for QC-failed but resellable inventory.

22. Glossary

Term	Definition
RMA	Return Merchandise Authorization — a unique identifier authorizing a product return
QC	Quality Check — inspection process to determine the condition of a returned item
SKU	Stock Keeping Unit — unique identifier for a distinct product/variant
PDP	Product Detail Page — the page displaying detailed product information
PLP	Product Listing Page — the page displaying a list/grid of products
COD	Cash on Delivery — a payment method where payment is collected at delivery
ATP	Available to Promise — real-time inventory availability for order/exchange fulfillment
GMV	Gross Merchandise Value — total value of merchandise sold through the platform
SLA	Service Level Agreement — a committed timeframe for process completion
OTP	One-Time Password — a temporary code used for identity verification
PII	Personally Identifiable Information — data that can identify a specific individual
Serial Returner	A customer exhibiting abnormally high return frequency, often indicative of policy abuse

23. Appendix

Reference Documents: ShopSphere Platform Modernization Program Charter (FY2026); ShopSphere Order Management Module BRD; ShopSphere Payment & Checkout Module BRD; ShopSphere Catalog & PDP Module BRD.

Related BRDs: BRD-EC-ORD-003 Order Management Module; BRD-EC-PAY-005 Payment & Checkout Module; BRD-EC-INV-009 Inventory & Warehouse Management Module.

Version	Date	Author	Change Description
0.1	05-Feb-2026	Vikram Singh	Initial draft circulated for internal review
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